

Executive Summary — aios-agent

Purpose

aios-agent is a lightweight SRE (Site Reliability Engineering) assistant that uses **native function calling** on a local Qwen model served by llama.cpp. It operates 100% offline in local mode, with an option for cloud models in cloud/hybrid mode. It runs Linux commands, reads and writes files, manages interactive processes, consults the web and learns from experience through procedural memory.

Main components

File	Function
<code>agent.py</code>	Function calling orchestrator, context compression and persistent sessions.
<code>tools.py</code>	Tool definitions and handlers (shell, read, write, web, git, MCP, processes).
<code>chat.py</code>	Interactive loop, config loading and mode selection.
<code>memory.py</code>	Persistent procedural memory per mode (ProcMEM).
<code>process.py</code>	Interactive process management via PTY.
<code>setup.py</code>	Initial configuration wizard (local/cloud/hybrid).
<code>aios-install</code>	Installer of the AIOS LFS ISO to hard disk.
<code>scripts/launch_llama.py</code>	llama-server launcher; does not create config, only starts in local/hybrid.
<code>scripts/firstboot.sh</code>	First-boot configuration.
<code>systemd/aios-llama.service</code>	systemd service for the model server (disabled at boot).
<code>systemd/aios-agent.service</code>	systemd service for the interactive agent (disabled).

What's new

1. Standard system paths

- Server: `/usr/local/bin/llama-server`
- Models: `/usr/local/share/aios/models/`
- Agent code: `/usr/local/bin/aios-agent/`
- Configuration: `~/.aios/config.yaml`
- API keys: `~/.aios/.env`

2. Install-to-disk (`aios-install`)

- GPT partitioning, ext4 formatting and copies the live system.
- Installs GRUB, generates `/etc/fstab` by UUID.
- Generates `~/.aios/config.yaml` for the aios user.

3. Refined systemd (lazy start)

- `aios-llama.service`: disabled at boot; `setup.py` enables and starts it only if the chosen mode is local or hybrid.
- `aios-agent.service`: disabled by default because it requires an interactive terminal.
- `sshd`: disabled in the ISO, with no fixed host keys; start it manually if remote access is required.

4. Passive `launch_llama.py`

- Does not create `~/.aios/config.yaml` by default.
- Exits cleanly with code 0 if no config exists or if mode is cloud.
- Starts the server only when mode is local or hybrid.

5. Reference to AIOS LFS

- This agent is included in the `aios-lfs` ISO: <https://github.com/ccarrillomanzanares/aios-lfs>.

Security

- Blocking of dangerous commands (`rm -rf /`, `dd`, `mkfs`, `fdisk`).
- Interactive confirmation for destructive commands.
- Write blocking on critical system paths.
- Restrictions on git operations (no `reset`, `rebase`, `merge`, `stash`, or branch deletion).

Requirements and deployment

- Python 3.10+, `requests`, `pyyaml`.
- llama.cpp server at `127.0.0.1:8083` for local/hybrid mode.
- To install to disk: run `aio-install` from the AIOS LFS live environment.

Project status

- Current version: **v0.16**
- Branch: `main`
- Last updated: July 24, 2026
- Repository: <https://github.com/ccarrillomanzanares/aio-agent>